Часть 1-2

Код получился более необычным из за моих ошибок, но все получилось в итоге

Код:

```
USE pr1;
GO

IF EXISTS (SELECT * FROM sys.tables WHERE name = 'OrderItems') DROP TABLE OrderItems;
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'Orders') DROP TABLE Orders;
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'BookCategories') DROP TABLE BookCategories;
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'Customers') DROP TABLE Customers;
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'Books') DROP TABLE Books;
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'Categories') DROP TABLE Categories;
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'Authors') DROP TABLE Authors;
GO

CREATE TABLE Authors (
    AuthorID INT PRIMARY KEY IDENTITY(1,1),
    FirstName NVARCHAR(50) NOT NULL,
    LastName NVARCHAR(50) NOT NULL,
    BirthYear INT NULL
);

CREATE TABLE Categories (
    CategoryID INT PRIMARY KEY IDENTITY(1,1),
    CategoryName NVARCHAR(100) NOT NULL UNIQUE,
    Description NVARCHAR(500) NULL
);

CREATE TABLE Books (
    BookID INT PRIMARY KEY IDENTITY(1,1),
    Title NVARCHAR(200) NOT NULL,
    AuthorID INT NOT NULL,
    ISBN NVARCHAR(13) UNIQUE,
    Price DECIMAL(10,2) NOT NULL,
    PublishedYear INT NULL,
    Pages INT NULL,
    FOREIGN KEY (AuthorID) REFERENCES Authors(AuthorID)
);

CREATE TABLE BookCategories (
    BookID INT NOT NULL,
    CategoryID INT NOT NULL,
    PRIMARY KEY (BookID, CategoryID),
    FOREIGN KEY (BookID) REFERENCES Books(BookID) ON DELETE CASCADE,
    FOREIGN KEY (CategoryID) REFERENCES Categories(CategoryID) ON DELETE CASCADE
);

CREATE TABLE Customers (
    CustomerID INT PRIMARY KEY IDENTITY(1,1),
    FirstName NVARCHAR(50) NOT NULL,
    LastName NVARCHAR(50) NOT NULL,
    Email NVARCHAR(100) UNIQUE NOT NULL,
    Phone NVARCHAR(20) NULL,
```

```
    Address NVARCHAR(200) NULL
);

CREATE TABLE Orders (
    OrderID INT PRIMARY KEY IDENTITY(1,1),
    CustomerID INT NOT NULL,
    OrderDate DATETIME NOT NULL DEFAULT GETDATE(),
    TotalAmount DECIMAL(10,2) NULL,
    Status NVARCHAR(20) NOT NULL DEFAULT 'New',
    FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

CREATE TABLE OrderItems (
    OrderItemID INT PRIMARY KEY IDENTITY(1,1),
    OrderID INT NOT NULL,
    BookID INT NOT NULL,
    Quantity INT NOT NULL,
    UnitPrice DECIMAL(10,2) NOT NULL,
    FOREIGN KEY (OrderID) REFERENCES Orders(OrderID) ON DELETE CASCADE,
    FOREIGN KEY (BookID) REFERENCES Books(BookID)
);
GO

INSERT INTO Authors (FirstName, LastName, BirthYear) VALUES
('Фёдор', 'Достоевский', 1821),
('Лев', 'Толстой', 1828),
('Антон', 'Чехов', 1860),
('Михаил', 'Булгаков', 1891);

INSERT INTO Categories (CategoryName, Description) VALUES
('Классика', 'Классическая литература'),
('Роман', 'Художественные романы'),
('Драма', 'Драматические произведения'),
('Философия', 'Философская литература');

INSERT INTO Books (Title, AuthorID, ISBN, Price, PublishedYear, Pages) VALUES
('Преступление и наказание', 1, '9785170000001', 450.00, 1866, 672),
('Война и мир', 2, '9785170000002', 890.00, 1869, 1225),
('Анна Каренина', 2, '9785170000003', 650.00, 1877, 864),
('Вишнёвый сад', 3, '9785170000004', 320.00, 1904, 128),
('Мастер и Маргарита', 4, '9785170000005', 520.00, 1967, 480),
('Братья Карамазовы', 1, '9785170000006', 780.00, 1880, 928);

INSERT INTO BookCategories (BookID, CategoryID) VALUES
(1, 1), (1, 3), (1, 4),
(2, 1), (2, 2),
(3, 1), (3, 2),
(4, 1), (4, 3),
(5, 1), (5, 2), (5, 4),
(6, 1), (6, 2), (6, 4);

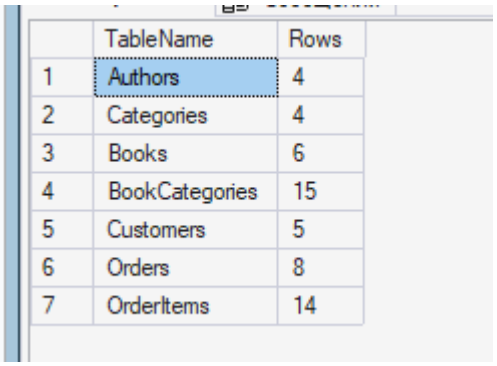
INSERT INTO Customers (FirstName, LastName, Email, Phone, Address) VALUES
('Иван', 'Петров', 'ivan@mail.ru', '+79001111111', 'Москва, ул. Ленина, 1'),
('Мария', 'Сидорова', 'maria@mail.ru', '+79002222222', 'СПб, Невский пр., 10'),
('Алексей', 'Смирнов', 'alex@mail.ru', '+79003333333', 'Казань, ул. Баумана, 5'),
('Елена', 'Козлова', 'elena@mail.ru', '+79004444444', 'Новосибирск, Красный пр., 20'),
('Дмитрий', 'Волков', 'dmitry@mail.ru', '+79005555555', 'Екатеринбург, ул. Мира, 15');
```

```
INSERT INTO Orders (CustomerID, OrderDate, TotalAmount, Status) VALUES
(1, '20240115', 450.00, 'Completed'),
(1, '20240220', 1340.00, 'Completed'),
(2, '20240310', 650.00, 'Processing'),
(3, '20240405', 1170.00, 'Completed'),
(4, '20240512', 520.00, 'New'),
(5, '20240618', 1760.00, 'Processing'),
(2, '20240722', 320.00, 'Cancelled'),
(3, '20240830', 1230.00, 'New');

INSERT INTO OrderItems (OrderID, BookID, Quantity, UnitPrice) VALUES
(1, 1, 1, 450.00),
(2, 2, 1, 890.00), (2, 1, 1, 450.00),
(3, 3, 1, 650.00),
(4, 5, 1, 520.00), (4, 4, 2, 320.00),
(5, 5, 1, 520.00),
(6, 6, 1, 780.00), (6, 2, 1, 890.00), (6, 1, 1, 450.00),
(7, 4, 1, 320.00),
(8, 3, 1, 650.00), (8, 5, 1, 520.00), (8, 4, 1, 320.00);
GO

SELECT 'Authors' AS TableName, COUNT(*) AS Rows FROM Authors
UNION ALL SELECT 'Categories', COUNT(*) FROM Categories
UNION ALL SELECT 'Books', COUNT(*) FROM Books
UNION ALL SELECT 'BookCategories', COUNT(*) FROM BookCategories
UNION ALL SELECT 'Customers', COUNT(*) FROM Customers
UNION ALL SELECT 'Orders', COUNT(*) FROM Orders
UNION ALL SELECT 'OrderItems', COUNT(*) FROM OrderItems;
GO
```

Скрин:



	TableName	Rows
1	Authors	4
2	Categories	4
3	Books	6
4	BookCategories	15
5	Customers	5
6	Orders	8
7	OrderItems	14